



Wellness Resort and Eco Spa

Sunnyside, FL



Ben, Caden, Emmy

1. Summary

Project Overview: The project is an upscale, wellness, and sustainability-focused resort and spa on 40 acres of undeveloped coastal lowland in Panama City Beach, Florida, situated between the Gulf of America and Lake Powell. The site is a high-risk flood zone.

Resiliency: Key threats include a severe flood risk (84/100), frequent Category 5 hurricanes, year-round wildfires, and sea level rise, which is predicted to cause regular flooding by 2060. Mitigation strategies include elevating all new infrastructure, using "living shorelines," implementing continuous load paths and impact-rated envelopes for hurricane resistance, and utilizing advanced HVAC filtration to address wildfire smoke.

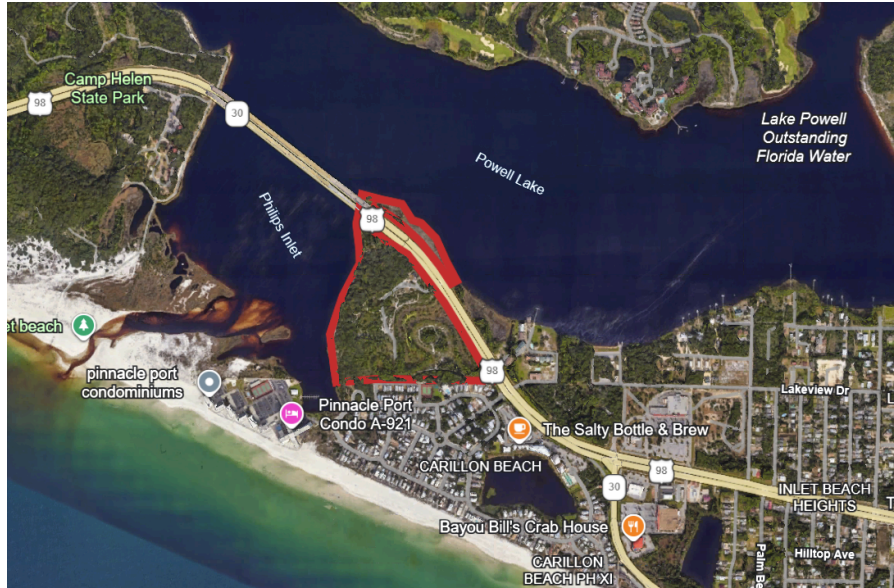
Energy & Buildings: Total building area is 376,708 sq. ft. with an annual energy usage of 15,688,896.22 kWh. A rooftop solar PV system on 161,435.33 sq. ft. of roof area will supply 20% of the annual building load (3,644,402.65 kWh yearly). The system has a 10-year payback period and results in net savings (NPV of -\$2,193,177.65) over 24 years.

Mobility & Transit: The selected Combination strategy promotes walking and uses an electric shuttle service and electric golf cart rentals. This option was chosen over the Base Case because it provides the maximum guest flexibility, autonomy, and wellbeing benefits, aligns with the high-end branding, and reduces emissions.

Water Systems: The design focuses on limiting impermeable surfaces and achieving a positive water balance for irrigation. It reclaims 100% of light gray water (3,292,942.40 gal) and collects rainwater (5,877,606 gal). These sources meet the total irrigation water demand (10,197,338 gal). The plan also uses three constructed wetlands and water-wise landscaping with native Florida plants, significantly reducing stormwater runoff compared to base conditions.

2. Introduction

Information about the site, basic stuff about the goals of the project.



The project site is located on the Florida Panhandle, about 10 minutes away from Panama City Beach, a major tourist attraction. The goal of the project is to create an upscale, resort and spa that is focused on wellness and sustainability. We will provide a premium, relaxing experience for guests who can enjoy amenities on site, and the nearby beaches and lake. Our site will feature native plant ecosystems, pools and spas, multiple lodging options, hiking trails, lake access and locally sourced dining. Our master plan provides our plans for resiliency issues, energy usage, transportation and water usage.

3. Current State

The coordinates 30°16'11.4"N 85°58'56.0"W place you in the western portion of Panama City Beach, Florida, specifically within the Carillon Beach and Sunnyside area. The land at this location is characterized by the following features: coastal lowlands, close proximity to water, sandy soil, 40 acres of undeveloped land off of Front Beach Road (Hwy 98).

The area is exceptionally flat, sitting at an elevation of approximately 5 to 10 feet above sea level. It is located on a narrow strip of land between the Gulf of America to the south and Lake Powell to the north. Lake Powell is one of Florida's rare coastal dune lakes, which are freshwater bodies located extremely close to the ocean. The natural soil is primarily composed of fine, white quartz sand. Native vegetation in the undeveloped pockets includes sea oats, scrub oaks, and sand pines. The land is serviced by Front Beach Road (Hwy 98), which is the primary artery for the Florida Panhandle's coastline. Because of its location on the Gulf Coast, the land is designated as a high-risk flood zone. It is subject to significant environmental regulations regarding beach nourishment and dune protection to mitigate the impact of tropical storms and hurricanes.

4. Specific Goals

Goal	Focus Areas
Energy & Buildings	Renewable energy, on-site generation, backup-power
Mobility & Transit	Promoting walking, electric shuttle, electric golf cart rentals
Water Systems	Stormwater management, permeable surfaces, water reclamation and collection, water-wise landscaping, wastewater management

5. Resiliency & Risk Mitigation

Key Resiliency Issues

The Resiliency & Adaptation Strategy for the Panama City Beach resort is critical due to the area's high vulnerability to acute and chronic environmental hazards. The site faces multiple significant threats, including a severe flood risk (84/100), frequent exposure to major hurricanes like Category 5 Hurricane Michael, year-round wildfire issues, and the long-term impact of sea level rise, which is predicted to cause regular flooding by 2060. This section is structured to first identify and analyze these Key Resiliency Issues, followed by a presentation of comprehensive Risk Mitigation strategies designed to adapt the resort's infrastructure and operations to withstand these challenges.

Flooding

Our largest risk comes from flooding. [Augurisk](#) notes a severe flood risk score of 84/100. The flood map for our location is shown below.



[FEMA](#) Flood risk map of our site

Green: 1% annual flood risk (would flood in 100 year storm), and a 26% chance of flooding over 30 years

Orange = 0.2% annual flood risk (would flood in 500 year storm), 5% chance of flooding over next 30 years

Risk finder is another tool to analyze flood risk. Their tool incorporates sea level rise into flood calculations. Assuming moderate sea level rise, risk finder predicts an “80% risk of at least one flood over 7 ft taking place between today and 2050 in the Panama City Beach area”. A 7 foot flood would decimate much of our site shown in blue below.



Result of a 7 foot flood from [Risk finder](#)

Hurricanes

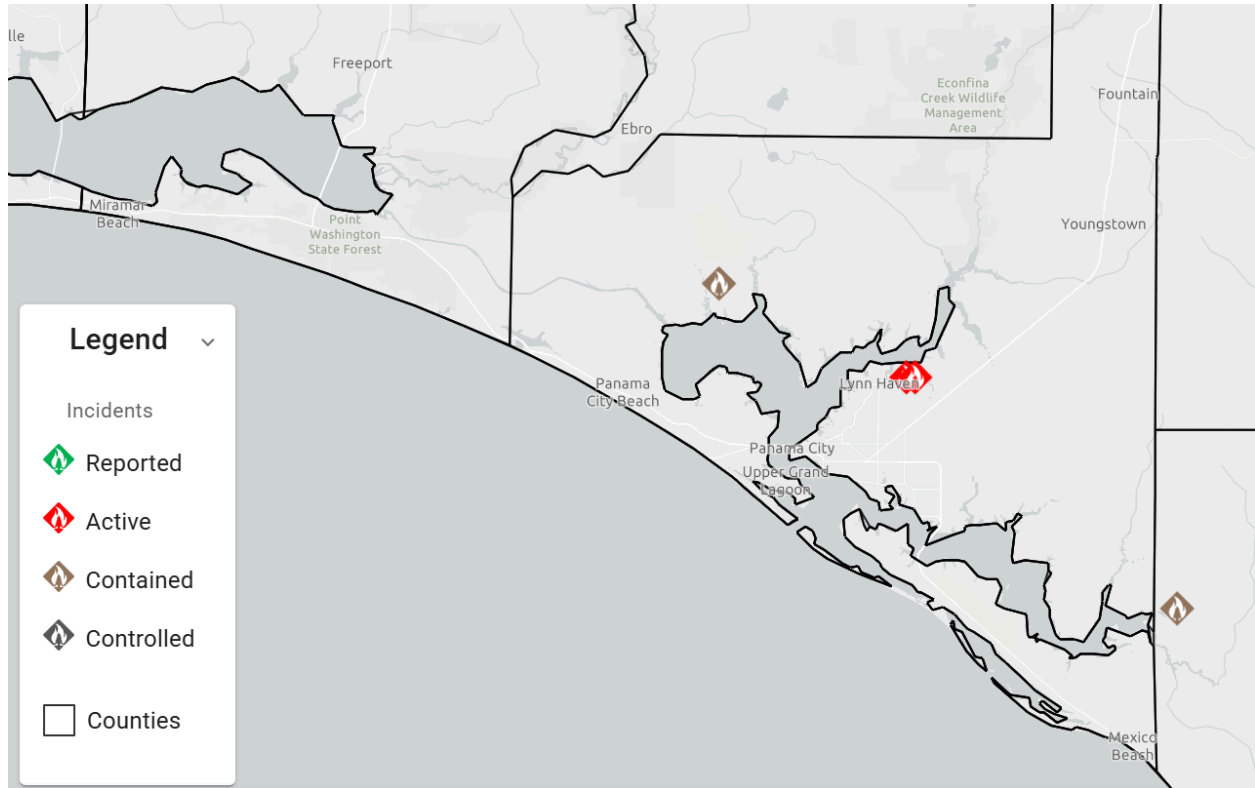
Hurricanes also pose a big threat to the Panama City Beach area. The most notable hurricane in recent memory was Hurricane Michael. Hurricane Michael was a category 5 hurricane and caused 74 deaths and \$25.5 billion dollars of property damage ([link](#)). Hurricanecity.com notes that that Panama City Beach area gets hit every 2.33 years and a direct major hurricane hits once every 19 years.



Damage from Hurricane Michael

Wildfires

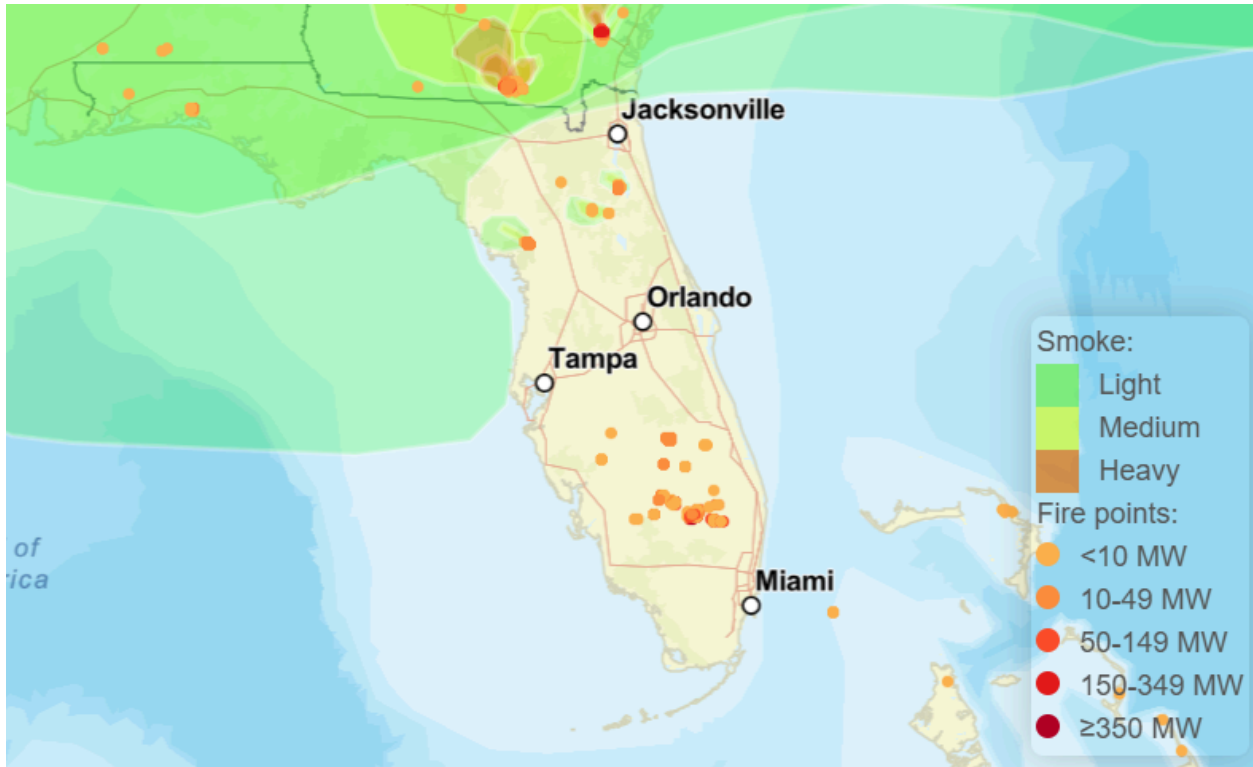
As of April 23, 2026, there are 134 active wildfires burning over 25,700 acres in Florida. In Bay County, where the resort is located, there are three fire incidents.



A map of fire incidents on April 23, 2026 from the Florida Fire Service

According to the Florida Department of Agriculture and Consumer Services, wildfires are a year-round issue. One fire management strategy the Florida Fire Service uses is prescribed burning. These controlled burns reduce the amount of flammable vegetation around the built environment.

Although prescribed burns serve as an effective fire management strategy, the smoke produced by this practice can have an effect on nearby communities. Not only is the property susceptible to wildfires, but the air may be contaminated by smoke. In fact, right now, April 23, 2026, the Florida panhandle is experiencing light smoke.



An image from the National Oceanic and Atmospheric Administration (NOAA) Hazard Mapping System Fire and Smoke Product Map on April 23, 2026

Sea Level Rise

Sea level rise is more of a long term issue but will definitely impact the future of our site. According to moderate to high protections of sea level rise our site would experience regular flooding in 2060 and much of our site would be underwater by 2100.



High sea level rise in 2060 (left) high sea level rise in 2100 (right)

Risk Mitigation

Flooding and Sea Level Rise

With predictions of regular flooding by 2060 and much of the site potentially underwater by 2100, a multi-layered defense is essential:

- **Elevated Infrastructure:** All new resort buildings should be constructed on elevated foundations, such as masonry piers, columns, or pilings, to allow storm surges to flow beneath the structures.
- **Utility Protection:** Critical mechanical and electrical systems, including HVAC units and electrical panels, must be placed on upper levels above the base flood elevation to prevent damage during inundation.
- **Shoreline Reinforcement:** Implement "living shorelines" by restoring coastal wetlands and dunes. This provides a natural buffer against erosion and storm surges while enhancing the eco-resort's aesthetic.
- **Deployable Barriers:** For existing low-lying areas, utilize temporary flood panels or deployable flood barriers (e.g., Tiger Dams) to provide short-term protection during high-water events.

Hurricane Resilience

To withstand Category 5 storms like Hurricane Michael, the resort will adopt these ideas:

- **Continuous Load Path:** Use hurricane straps and clips to create a secure, reinforced connection from the roof to the walls and down to the foundation.
- **Impact-Rated Envelopes:** Install impact-resistant glass and heavy-duty reinforced frames for all windows and doors to prevent the building envelope from breaching during high-wind events.
- **Hardened Roofing:** Prioritize metal roofing or concrete tiles secured with a high density of fasteners and advanced adhesives to prevent roof lift.

Wildfire and Air Quality

As wildfires are a year-round issue in Florida, the resort must manage both fire risk and the resulting smoke:

- **Defensible Space:** Reduce flammable vegetation around the built environment by maintaining a prescribed burning schedule in coordination with the Florida Fire Service.

- Advanced Filtration: To mitigate "light smoke" conditions common in the panhandle, resort HVAC systems will be equipped with HEPA-grade air filtration and sensors to automatically switch to internal air recirculation when outdoor air quality (AQI) drops.
 - Real-time Monitoring: Integrate tools like the NOAA Smoke Forecast and AIRNow to proactively notify guests of air quality concerns.
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6. Implementation

Energy & Buildings

Each building included in the master plan was categorized and its area was measured (Table 1). Building energy codes from the United States Department of Energy were used to calculate energy usage of each building (Table 2).

- Key
- (R) Resort
Amenities (first floor) – approx. 20,000 sqf
60 Keys
 - (I) Medical Institute
approx. 106,000 sqf
 - (M) Lakefront Manor
8 units
 - (LM) Lakefront Maison
2 buildings with 3 units each (total 6 units)
 - (GM) Garden Maison
4 buildings with 4-6 units each (total 20 units)
 - (PT1) Pied-a-terre 1
Amenities (first floor) – approx. 31,000 sqf
16 or 18 units (2nd and 3rd floors)
6 Penthouse units
 - (PT2) Pied-a-terre 2
Amenities (Institute) – approx. 6,500 sqf
21-24 units (2nd and 3rd floor)
6 – 7 Penthouse units
Back of house approx. 13,000 sqf
 - (PT3) Pied-a-terre 3
Amenities – approx. 12,500 sqf
41 to 43 units (2nd and 3rd floor)
6 to 7 Penthouse units
 - (PT4) Pied-a-terre 4
15 – 20 units
3 – 4 Penthouse units
 - (PM) Peninsula Maisonette
9 – 11 units
TBC upon receipt of new wetland survey
 - (A) Back of House
 - (B) Gatehouse
 - (C) Boardwalk
 - (D) Gardens
 - Existing Trees
 - Proposed Trees
 - Wetland Buffer 2001
 - Wetland Buffer 2016
 - Indicative Extent of Basement (to be confirmed)



Eco Spa and Wellness Resort Master Plan

Category	Building Type	Quantity	Number of Floors	Unit Area (sq. ft.)	Total Area (sq. ft.)	Roof Area (sq. ft.)
Resort	Lodging - Large Hotel	1	1	20,000	20,000	20,000
Medical Institute	Healthcare - Outpatient	1	2	106,000	106,000	53,000
Lakefront Manoir	Apartment - Mid Rise	8	2	2,500	20,000	10,000
Lakefront Maison	Apartment - Mid Rise	2	2	2,500	5,000	2,500
Garden Maison	Apartment - Mid Rise	2	1	12,600	25,200	12,600
Pied-a-terre 1	Apartment - Mid Rise	1	4	31,000	124,000	31,000
Pied-a-terre 2	Apartment - Mid Rise	1	4	24,690	24,690	6,173
Pied-a-terre 3	Apartment - Mid Rise	1	4	25,318	25,318	6,330
Pied-a-terre 4	Apartment - Mid Rise	1	3	10,000	10,000	3,333
Peninsula Maisonette	Apartment - Mid Rise	11	1	1,500	16,500	16,500
Parking Structures	Parking	1	2	68,400	136,800	68,400

Table 1: Building Categories and Areas

Category	Source EUI (kBtu/ft2-year)	Total Area (sq. ft.)	Energy usage (kBtu/year)	Energy Usage (kWh/year)
Lodging - Large Hotel	164.8	20,000	3,296,000	965,962
Healthcare - Outpatient	228.9	106,000	24,263,400	7,110,899
Apartment - Mid Rise	103.6	250,708	25,973,349	7,612,035
Totals		376,708	53,532,749	15,688,896.22

Table 2: Energy Usage

Florida's sunny weather makes solar energy a reliable choice for energy. Solar panels will be installed on all the rooftops on-site. This will not interfere with the aesthetics of the resort. Table 3 displays the yearly PV generation. The proposed system provides

20% of the total annual building load (Figure 2). Although not all of the energy needed can be supplied by the panels, the net present value (NPV) over 24 years is -\$2,193,177.65, indicating net savings. The payback period is 10 years.

kWh/year/kw	kW/sqft	kWh/year/sqft	System size (kW)	Total roof area (sqft)	Area roof used for pv (sqft)	Yearly PV generation (kWh)	Yearly building load (kW)
1,505.00	0.015	22.58	2421.53	161,435.33	161,435.33	3,644,402.65	15,688,896.22

Table 3: PV Calculation

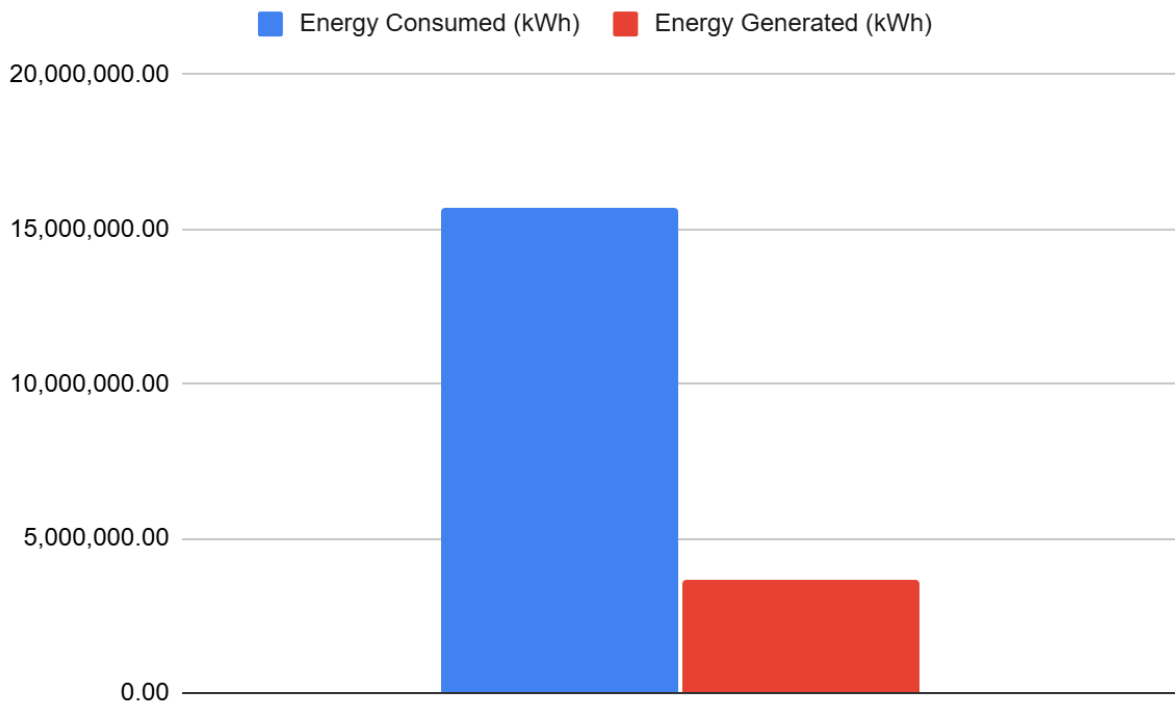


Figure 1: Building Load and PV Energy Generation

Mobility & Transit

The base case for our transportation strategy is everybody driving their personal vehicles to and from the resort. That also includes driving through the resort to their dwellings, medical institute, etc. The base case also incorporates 15 golf carts for staff maintenance and grounds crew priced at \$10,000 each.

We came up with 3 strategies and analyzed the combinations of all of them.

Our first alternative is to encourage walking by having all resort guests parked in a parking garage and guests walk around the resort. The design strategies to encourage walking revolve around making walking more comfortable and making the walk an 'experience'. In order to improve guest comfort, we need to increase shading by planting native trees such as Live Oak, Longleaf Pine, and Bald Cypress. Each tree will be labeled, like in a botanical garden making the walk a learning experience. Alongside the paths, we can incorporate stands that sell fresh fruits, and drinks. Furthermore, adding benches and places for resting encourages guests to take a break and enjoy the scenery. The walking paths will stay away from the road to decrease noise and pollution.

The second alternative is an electric shuttle service that makes rounds of a 2 mile loop around the property. We will have four shuttles that arrive every 5 minutes during peak traffic times such as morning and dinner time. During the rest of the day, we will have two shuttles, meaning one will arrive every 10 minutes. These shuttles will be all electric and can hold 15 passengers per shuttle. The shuttles would cost around \$115,000 each.

The third alternative is having electric golf carts for rent. These allow guests to make trips around the resort. There would be a private golf cart for each of the manors, maisons and maisonettes, meaning 45 golf carts for these more private dwellings. Each unit would have a place to park the golf cart. For the denser dwellings (pied-a-terre and resort) we assumed that the golf carts would be shared between the units with 1 golf cart for every 3 units on a renting basis. With around 180 guests in these higher density areas, that gives 60 golf carts. Giving us a total of 105 golf carts. These could be located in a shared underground parking lot to reduce space requirements and improve aesthetics. The golf carts are priced at \$10,000 each.

Finally, the fourth combines all three of the previous designs. Walking is encouraged when possible but shuttles and golf carts are also available. This design assumes 2 shuttles during rush times and 1 shuttle during down times but also provides a golf cart rental service of 50 golf carts. This strategy allows for maximum flexibility for guests and doesn't overcrowd the road with many golf carts zipping around.

The plan with all of these options is to have as many amenities on site as possible to reduce off-site trips.

Assumptions

Assumed an average of 2 people/unit for resort and pied-a-terre and 4 people/unit for manoir and maisons giving an estimated guest count of 466 people at peak capacity. Therefore, we assumed a total of 466 guests per week/ We also assumed that there is 1 car for every 2 people. We assumed that 10% of guests drive an EV ([DOE](#)). We assumed the average gas car emits 0.88 lbs of CO₂ per mile driven data from [EPA](#). and 0.35 kWh/mile for EV

We also assumed an average of 75 miles between the guests home and the resort, resulting in a baseline of 150 miles per car for external trips. For this eco-resort, we believe it is safe to assume that it will be semi-local, which we determined to be about an hour, to an hour and a half drive.

Assumed golf carts require 0.15kWh/mile and have annual O&M costs of \$1000

For guests that want to eat at a restaurant or get groceries, we assumed that they would drive an average of 10 miles for their entire stay, or 2 miles a day for a 5 day stay. That would include a mile per day of internal driving.

Assumed energy cost for commercial business in PCB is \$0.10/kWh ([BCEC](#))

We assumed the shuttle would travel around 10 mph and have 5 stops throughout the property. We assumed each stop would take about 1.5 minutes giving us about 20 minutes per cycle. If we are running shuttles constantly, we could have increased shuttle usage during peak times (7-10 am and 6-9pm). The shuttle will have an efficiency of 1.5 kWh per mile. The annual O&M cost for shuttles \$300,000.

We assumed the social cost of carbon is \$190 per ton of CO₂ ([RFF](#)) .

Assumed price to make parking lot \$5000 per stall for non EV stalls and \$10,000 for EV stalls with 20% of stalls having EV capability.

Assumed Landscaping walking experience installation cost \$1,000,000 and annual O&M \$100,000. Costs encapsulate tree planting, landscaping, path construction, labeling and kiosk construction. Assumed the trees planted sequester 48lbs/carbon/year and 350 trees planted for walking and combination case.

Analysis

Costs and NPV were calculated (Table 1). The Base Case had the lowest NPV with the majority of the costs coming from installing a parking lot and carbon cost. The Shuttle had the highest NPV with the majority of costs coming from yearly O&M.

Although the base case appears to have the lowest NPV, it is important to note the ecological, environmental, and wellbeing benefits associated with the walking and combination strategies. Allowing guests to have easier access to amenities without needing to drive allows for greater comfort and wellbeing and fits well with the high-end branding of the resort. Therefore the base case isn't an option. Furthermore, the combination option provides guests with the greatest freedom and autonomy while still reducing emissions.

	Installation Cost (\$)	Energy Cost (\$/yr)	Carbon Cost (\$/yr)	O&M (\$/yr)	Total Cost (\$/yr)	NPV (30yr)
Base case	1,398,000.00	16,502.13	51,190.29	13,980.00	81,672.42	2,653,505.25
Walking	2,398,000.00	0.00	-1,596.00	100,000.00	98,404.00	3,910,710.67
Shuttle only	1,858,000.00	10,117.80	0.00	300,000.00	310,117.80	6,625,270.69
Golf cart only	3,498,000.00	2,213.27	0.00	105,000.00	107,213.27	5,146,130.72
Combination	3,128,000.00	6,112.84	-1,596.00	202,500.00	207,016.84	6,310,356.20

Table 3: Comparison of Transportation Options

Water Systems

We can implement 3 constructed wetlands for storm water total area of constructed wetland is 4,200 sq. ft. Impermeable surfaces will be limited as paths will be gravel or permeable brick paved paths. We will also reclaim 100% of light gray water produced, and collect rain water from all rooftops. The combination of those sources will be used for irrigation. The remaining water demand will be met by purchasing municipal water.

In order to reduce our water usage for irrigation purposes we opted to create a native Florida plant ecosystem. These plants are adapted to the climate of Florida and therefore do not need to be watered nearly as much. We will use plants from this list from the University of Florida. These plants are not only less water intensive but also support local wildlife which plays strongly into our narrative of a sustainable eco resort.

For the black water systems, we are sending it through the local municipal sewage system to avoid the risk of water events causing contamination to the local ecosystem. For the dark graywater, we will be sending it through a constructed wetland, which will need a surface area of 390 square meters.

Assumptions

Assuming only $\frac{1}{3}$ of the property will be irrigated (528,000 sq ft). We came to that conclusion because we assumed that $\frac{1}{3}$ of the property is paved, and that the other third is natural and engineered wetlands. We also assume that irrigation will only need to happen every other day for 6 months out of the year, since the dry month is winter, where it is colder.

For the native plant ecosystem, we assumed a smart drip irrigation system that detects soil moisture, mulch planting beds, and plants already well established.

Assuming the wetlands are classified as infiltration basins on the EPA national stormwater calculator.

For the constructed wetland, a lot of assumptions had to be made. The state of Florida does not have clear regulations for calculating the surface area needed for the wetland. We assumed effluent BOD of 30 mg/l and K_{BOD} of 0.15 m/d. These numbers came from the example problem in the constructed wetlands manual.

Analysis

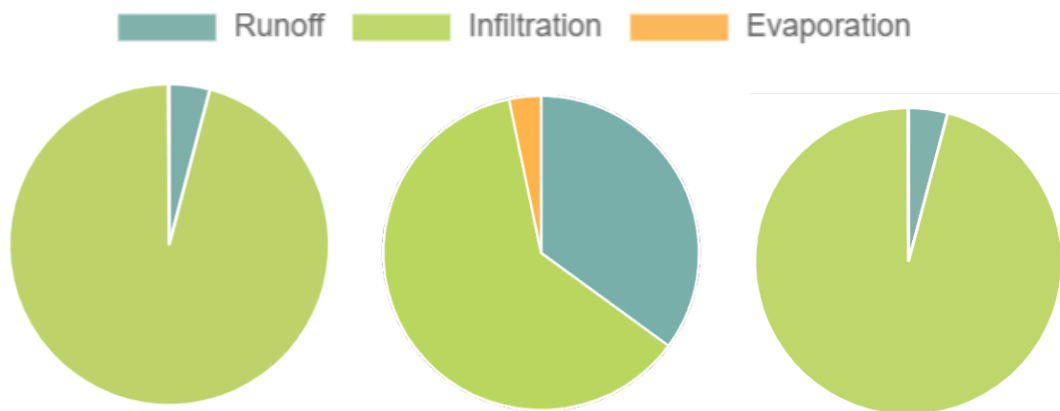
[Water Strategy - Google Sheets](#)

Water demand	10,197,338 gal
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Rainwater collection	5,877,606 gal
Gray water reuse	3,292,942.40 gal
Total water	10,367,325.75 gal
Water cost	\$42,817.06

Table 4: Water balance summary

Stormwater was analyzed using the EPA's National Stormwater Calculator. The base case, which featured no low-impact development (LID) features, has approximately twice the amount of runoff as our design, which features rainwater harvesting, infiltration basins (coming from the natural wetlands), and permeable pavement.



Stormwater summaries for the pre development conditions (left) base case (middle) and LID case (right)

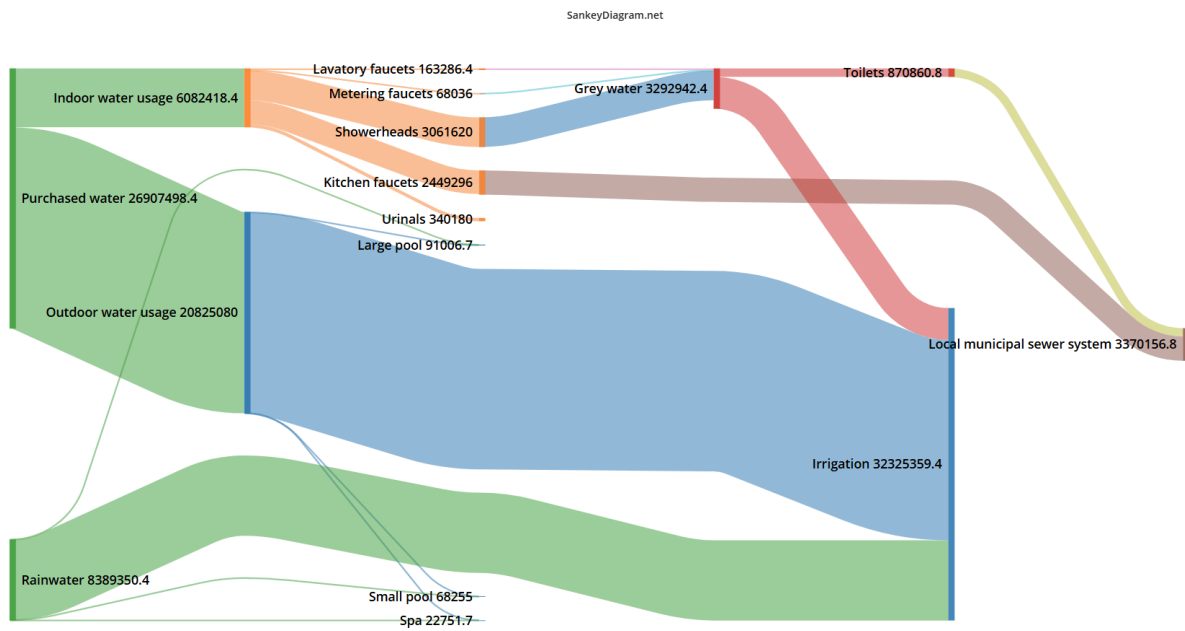


Figure 2: Sankey Diagram

https://www.energycodes.gov/sites/default/files/2024-02/Standard_90.1-2022_Final_Determination_TSD.pdf